COOLSHADE · USER REQUIREMENTS

CoolShade user requirements

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DOCUMENT CONTROL

Document ID	CSH-URS-001	Version	0.1
Status	Draft	Date	2026-10-30
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/coolshade		
Source	docs/03a-user-requirements.md @ ae091ec		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-30	Amish Chadha	First issue in the kit 1.9.1 upgrade, written from the users and stakeholders in CSH-PRB-001 and the 2026-10-02 market lane decision; used to score the concept options (CSH-OPT-001); every need traced to a product requirement in CSH-PRS-001; a desk draft until a co-design partner reviews it

CoolShade user requirements

What the people who wait, sell, run and fund a public cooling shade need, written from their side. The fifteen needs come from the users and stakeholders in CSH-PRB-001 Table 1 (transit riders, market traders and shoppers, people in queues, outdoor workers, the city, transit or market operator, community groups and schools) and from the first site decided on 2026-10-02, a market lane in a hot, dry city (CSH-DEC-001). They are used to score the concept options (CSH-OPT-001) and each one has at least one product requirement in CSH-PRS-001 (links in CSH-RTM-001). They remain a desk draft: no co-design session has been held and no partner has agreed to anything. Method: STANDARDS section 22.

Table 1. User requirements.

ID	Category	Title	Need	Priority	Source
UR-001	Performance	Shade from the sun	I can stand out of direct sun, with the heat from the sun and hot surfaces cut, while I wait	High	Transit riders, queues, outdoor workers (PRB Table 1)
UR-002	Performance	Cooler air	The air where I stand feels cooler than the street, also when it is still	High	Transit riders, market traders
UR-003	Usability	I stay dry	My skin, clothes and goods do not get wet, and the ground does not get slippery	High	Transit riders, market traders
UR-004	Usability	Room for everyone	There is room for a group to stand or sit, and for a wheelchair, and I do not hit my head	High	Queues, older adults, wheelchair users (PRB)
UR-005	Performance	Works only when needed	It cools when it is hot and someone is there, and stops otherwise, so water and power are not wasted	Medium	Operator
UR-006	Deployment and assembly	No power connection	It works where there is no mains power	High	Operator, market lane (PRB constraints)
UR-007	Safety	Safe water	The mist and the water behind it cannot make me ill	High	All users; city and market operator
UR-008	Safety	Stands up to wind and weather	It does not fall, fly or collapse in a gust or a storm	High	Operator, everyone nearby
UR-009	Safety	Safe electrics	I cannot get a shock or start a fire, and nothing is hot or sharp where I can reach	High	All users, operator
UR-010	Safety	Privacy	It does not watch, film or record me	Medium	Transit riders, market traders
UR-011	Maintenance	Easy to keep running	Refilling, cleaning and clearing the nozzles take little time and few tools	Medium	Operator, community groups
UR-012	Deployment and assembly	The community can build it	A community workshop or school can build it from common parts and tools	Medium	Community groups and schools
UR-013	Logistics	It can be moved	Two people can move it to another site	Medium	Operator
UR-014	Cost	Affordable	It costs no more than a market or a district can raise for one site	Medium	Operator, NGO
UR-015	Environment and durability	Uses little water	A tank lasts at least a day and a design day uses little water	Medium	Operator, city water authority

Categories: deployment and assembly, performance, usability, safety, environment and durability, logistics, maintenance, cost, regulatory. Priorities: High, Medium, Low.